International Journal of Forecasting 41 (2025) 424–439

Contents lists available at ScienceDirect

International Journal of Forecasting

journal homepage: www.elsevier.com/locate/ijforecast

An overview of the effects of algorithm use on judgmental biases affecting forecasting

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ARTICLE INFO

Keywords:
Judgmental biases
Algorithm use
Judgmental accuracy
Judgmental forecasting
Judgmental adjustment
Review article

ABSTRACT

In the realm of forecasting, judgmental biases often hinder efficiency and accuracy. Algorithms present a promising avenue for decision makers to enhance their forecasting performance. In this overview, we scrutinized the occurrence of the most relevant judgmental biases affecting forecasting across 162 papers, drawing from four recent reviews and papers published in forecasting journals, specifically focusing on the use of algorithms. Thirty-three of the 162 papers (20.4%) at least briefly mentioned one of twelve judgmental biases affecting forecasting. Our comprehensive analysis suggests that algorithms can potentially mitigate the adverse impacts of biases inherent in human judgment related to forecasting. Furthermore, these algorithms can leverage biases as an advantage, enhancing forecast accuracy. Intriguing revelations have surfaced, focusing mainly on four biases. By providing timely, relevant, well-performing, and consistent algorithmic advice, people can be effectively influenced to improve their forecasts, considering anchoring, availability, inconsistency, and confirmation bias. The findings highlight the gaps in the current research landscape and provide recommendations for practitioners. They also lay the groundwork for future studies on utilizing algorithms (e.g., large language models) and overcoming judgmental biases to improve forecasting performance.

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1. Introduction

Forecasting plays a crucial role in our daily decision-making, and its impact greatly depends on its accuracy and precision (Makridakis et al., 2020). Despite the expansion and sophistication of forecasting methods (Petropoulos et al., 2022), human judgmental biases continue to influence the accuracy of forecasts significantly. As Hahn and Harris (2014, p. 42) mentioned, "the word 'bias' means many things to many people". The term "bias" serves as an umbrella for any psychological factor that may systematically impede the accuracy of a human forecaster (Hahn & Harris, 2014; Kahneman, 2000). Such factors can, for example, lead to insufficient utilization of algorithms, thereby yielding detrimental outcomes despite algorithms' typically superior predictive capabilities.

Nearly 70 years ago, Meehl (1954) conducted a groundbreaking review showcasing that algorithms regularly outperform humans in predictive performance. Subsequent studies have consistently confirmed this finding (Agrawal et al., 2018; Armstrong, 2001; Dawes, 1979; Grove et al., 2000; Kaufmann & Wittmann, 2016). Considering that the concept of an algorithm has evolved, encompassing a wide range of applications that perform tasks, the term "algorithm" is used nowadays as an umbrella for any kind of mechanical decision aid. Despite

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https://doi.org/10.1016/j.ijforecast.2024.09.007
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the widespread recognition of algorithms' ability to enhance decision-making outcomes, extensive research suggests that decision makers often are reluctant to rely on algorithms—a phenomenon observed when individuals witness algorithmic mistakes. This is called algorithm aversion (Dietvorst et al., 2015).

In recent years, there has been a growing interest in understanding the use and adoption of algorithms, particularly in the context of forecasting. In the swiftly advancing landscape of advanced algorithms such as generative artificial intelligence and large language models (LLMs), firms and individuals stand to gain significantly by fully leveraging their potential benefits (Haefner et al., 2023). Several studies have explored the underlying factors tied to algorithm use (Chacon et al., 2024; Dietvorst et al., 2015; Logg et al., 2019). While some studies on the topic have mentioned specific biases, such as anchoring and confirmation biases (Feng & Gao, 2020; Scherer et al., 2015), no systematic review has been proposed that focuses on how algorithms might help improve forecasting by considering decision makers' judgmental biases. Hence, we provide such an overview.

In this comprehensive overview, we consider the primary judgmental biases (twelve in total; see Appendix 1[1] for details) that impact forecasting, as outlined by Makridakis and Bakas (2016). Subsequently, we analyze the presence and extent of these twelve biases across 162 studies included in four recent reviews (Burton et al., 2020; Jussupow et al., 2020; Kaufmann et al., 2023; Mahmud et al., 2022) and studies published in forecasting journals focusing on the utilization of algorithms. We classify each study based on the depth of its coverage of the bias, distinguishing between those that address the topic effectively and those that only mention it briefly. Papers that use the term in unrelated contexts are excluded. Emphasizing the relevance of review research (Chalmers et al., 2002; Kunisch et al., 2023), we offer a current overview of how employing algorithms can mitigate the adverse effects of biases in forecasting or enhance forecasting performance by strategically leveraging biases to our advantage. Hence, we not only cover a previously unresearched topic but also consider all prior relevant reviews, making our dataset more complete than previous ones.[2]

In addition to traditional overviews and previous reviews on the topic, we used a text-mining methodology to support our coding procedure (Ananiadou et al., 2009; Thomas et al., 2011). Hence, we could explore this area, pinpoint existing research gaps, and formulate conclusions. This approach allows us to propose recommendations and guidelines for future research.

This study's primary contribution and innovation lie in its exploration, findings, and proposed future research paths regarding the role algorithms can play in considering biases in forecasting. While previous literature has separately examined the use of algorithms or judgmental forecasting, to the best of our knowledge, this research represents the inaugural effort to amalgamate the capabilities of algorithms in enhancing forecast accuracy by addressing judgmental biases.

Our findings showed that four key biases—anchoring, availability, inconsistency, and confirmation—are more addressed than other biases in the algorithm use literature. First, algorithms have the potential to counteract anchoring bias by providing precise and impartial recommendations at the beginning of the forecasting process. This approach decreases the dependence on distorted information and repurposes anchoring bias to our benefit, thereby enhancing the quality of decision-making in forecasting. Second, they broaden forecasters' perspectives against the availability heuristic by supplying relevant data, overcoming the limitations of human memory. Third, algorithms may ensure consistent decision-making, unlike the variable nature of human choices, thus addressing inconsistency. Finally, confirmation bias can be leveraged by pre-selecting effective algorithms, boosting confidence in their recommendations, and making them a reliable advice source alternative. The other eight biases Makridakis and Bakas (2016) outlined were either briefly addressed in the literature or, in some cases, entirely overlooked.

2. Theoretical background

2.1. Forecasting

We depend on forecasting daily—weather forecasts, stock forecasts, and health-related forecasts, among others. Forecasts also impact our planning and decision-making. For example, if we expect stock markets to increase, we may spend our savings more easily than if the economic forecast is pessimistic. Today, forecasting is performed by diverse methods, and many tools help experts improve their ability to forecast accurately (Petropoulos et al., 2022).

Forecasting advice is crucial for decision-making processes across various domains, such as business, environmental, and agricultural settings. For example, meteorologists use supercomputers to come up with their weather forecasts. In agriculture forecasting, optimization results can offer valuable advice to local farmers on planting and irrigation practices (Shan et al., 2019). In aviation, accurate forecasts of phenomena such as volcanic ash clouds are crucial for aviation safety during such events (Fu et al., 2016). In fisheries management, forecasts must balance uncertainty with practicality to avoid harmful consequences such as overly optimistic catch advice (Van Beveren et al., 2021).

Although forecasting advice is given in many domains, multiple factors influence advice acceptance, such as trust in the forecast source, individual decision preferences, and the nature of the advice (Mayer et al., 2023). The type of information and the quality of advice can significantly impact decision makers' choices, influencing which alternative is selected (Harvey, 2020; Lackes et al., 2020). The perceived credibility of the advice giver plays a significant role in how the advice is weighted and accepted (Önkal et al., 2017). From the decision maker's

[1] Appendices are available at OSF: https://osf.io/38ypz.

[2] See Kaufmann et al. (2023) for a description of the overlapping studies.

perspective, judgmental biases—understood as psychological factors that systematically hinder the accuracy of human forecasters—affect both their forecasting process and overall performance.

Forecasting is affected by different human biases that can significantly affect the accuracy and reliability of predictions (Eroglu & Croxton, 2010; Lawrence et al., 2006; Wright et al., 1996). Moreover, since Kahneman and Tversky's famous research on biases (Tversky & Kahneman, 1974), the spotlight has been on biases, and various authors have identified and discussed a range of them in different contexts unrelated to forecasting—for instance, overviews of biases focusing on environmental transformations, strategic planning, norm violations, or beliefs guiding information processing (Acciarini et al., 2020; Barnes, 1984; Jussim, 2012; Krueger & Funder, 2004; Oeberst & Imhoff, 2023; Pronin & Hazel, 2023; for details we refer to Appendix 2).

In the context of probabilistic forecasting, biases can arise. Understanding these biases and their implications is essential for improving forecasting performance and decision-making processes. In their comprehensive review, Petropoulos et al. (2022) acknowledge the presence of certain biases. However, their primary aim is to explore various models, methodologies, principles, and strategies relevant to forecasting extensively. There is limited research on biases in judgmental probabilistic forecasting contexts (Lawrence et al., 2006; Makridakis et al., 2020). Focusing on our primary interest—forecasting—the studies by Makridakis (1988) and Makridakis and Bakas (2016) are unique in combining the topics of forecasting and biases (see Appendix 2). They identify the twelve most relevant biases affecting forecasting, such as anchoring, conservatism, and recency (see Table 6 in Makridakis & Bakas, 2016). Therefore we choose Makridakis and Bakas' (2016) list to conduct the present overview.

As previously discussed, algorithms usually offer predictive capabilities superior to those of human forecasts. This advantage presents a compelling opportunity to enhance forecasting accuracy. Therefore, it is essential to conduct further research to expand our knowledge of algorithms' potential. This includes understanding how they can help reduce human biases or leverage them to our advantage in forecasting scenarios. Prior to addressing this research gap, we provide an overview of existing research on algorithm use.

2.2. The use of algorithm advice

The understanding of what constitutes an algorithm has changed over time. Initially, an algorithm could be understood as a finite, well-defined set of instructions or rules designed to perform a specific task or solve a particular problem. In recent years, however, traditional algorithms have undergone profound transformations propelled by digitalization, ushering in the era of artificial intelligence (AI). A pertinent illustration of this shift is the ubiquitous use of smartphones, which now serve as indispensable tools, using algorithms to guide us through the intricacies of our daily lives. Therefore, the concept of an algorithm has evolved and now encompasses a wide range of applications and complexities across various fields.

Focusing on how algorithms can aid in reducing human bias in forecasting is of paramount importance. In today's landscape, forecasting is conducted using a variety of methods and tools (algorithms), enhancing experts' capacity for accurate predictions. For example, generative AI and LLMs can refine code generation, enhance user interfaces, and bolster the interpretability of forecasting software (Trovero & Potamitis, 2023). Consequently, additional insights—whether from an expert or an algorithm—frequently serve as a secondary check in any forecasting endeavor. This underscores the critical role algorithms can play in both enhancing accuracy and safeguarding against biases.

Algorithmic advice refers to guidance or recommendations provided by programmed algorithms, increasingly utilizing AI (Wexler & Oberlander, 2023). This form of advice is distinct from human advice as it is based on automated professional knowledge managed by AI, making it a unique blend of technology and expertise. Algorithmic advice represents a novel approach to providing guidance and recommendations by leveraging programmed algorithms and AI. Its unique characteristics, effectiveness, and influence on decision-making processes distinguish it from traditional human advice, shaping the landscape of professional advice and decision-support systems.

Due to the importance of algorithm use, many studies have been conducted to check whether expert advice or algorithm advice is preferred as an additional advice source by experts, also leading to several overviews within the field (Burton et al., 2020; Jussupow et al., 2020; Kaufmann et al., 2023; Mahmud et al., 2022). The first review of the four within the field was published in 2020 and focuses on studies reporting algorithm aversion (Dietvorst et al., 2015). Burton et al. (2020) proposed causes and category-dependent solutions to overcome algorithm aversion in five categories: expectations and expertise, decision autonomy, incentivization, cognitive compatibility, and divergent rationalities (for details, see also (Burton et al., 2023)).

The follow-up review by Jussupow et al. (2020) aims to supplement Burton's review, focusing on algorithmic and human agent characteristics and including algorithm appreciation (Logg et al., 2019) studies. They revealed that algorithm agency, performance, capabilities, human involvement, human advisors' expertise, and social distance influence algorithm use. Further reviews by Mahmud et al. (2022) and Kaufmann et al. (2023) supplement Jussupow et al.'s (2020) review, which also focuses on user characteristics, task characteristics, and the characteristics of their interactions.

These reviews reveal that algorithm use and adoption are complex and influenced by various factors and contexts. Consequently, despite their potential to enhance forecasting, algorithms are likely underutilized. Hence, the potential impact of algorithm advice may not be fulfilled in forecasting tasks.

2.3. Research gaps on algorithm advice use: Biases

2.3.1. Algorithms and human biases

Currently, biases found from advice sources (i.e., algorithms) are a hotly discussed topic. In the context of algorithms, biases can be understood as factors that may undermine the accuracy of algorithmic forecasts. For example, Rovatsos et al. (2019) revealed biased algorithm advice in four case studies from different application fields—financial services, local government, crime and justice, and recruitment. Bias in algorithms can manifest at any point within the AI lifecycle, as De Silva and Alahakoon (2022) noted. They may originate from the data itself, the interaction between the data and the algorithm, or directly from the developers who program the algorithms. The presence of biases in these advice sources is well documented and has prompted significant political responses, including a recent unanimous United Nations resolution on AI.[3] Although biased algorithm advice is important, it is not within the scope of our review. Our focus is on biases in human forecasting, not on biases in algorithmic advice; for readers who are interested in the topic, please read Mosqueira-Rey et al. (2023). In the following, we do not link bias to algorithmic advice, to avoid confusing readers.

We believe that we must urgently focus on human users (see also Schwartz et al., 2022) and their judgmental biases when interacting with algorithm advice. Even when algorithmic advice is free from bias and improves human forecasts, are human users also free from biases when utilizing this advice? Specifically, do any of the twelve judgmental biases affecting forecasting identified by Makridakis and Bakas (2016) still impact their forecasts?

Although substantial evidence suggests that advice from algorithms tends to be more accurate than human judgment, it is also well documented that forecasts derived from algorithms frequently undergo human adjustment (referenced by De Baets & Harvey, 2020; Fildes & Goodwin, 2007; Fildes et al., 2009; Fildes & Petropoulos, 2015; Önkal et al., 2023, p. 247). Echoing De Baets and Harvey (2023, p. 273), we propose that forecasting adjustments are prevalent throughout the AI lifecycle and in its application by end users. These adjustments often incorporate factors beyond the algorithm's original parameters, such as contextual elements in its recommendations. This phenomenon, known as the "broken leg" phenomenon (Salzinger, 2005), highlights the necessity for human experts to determine whether a specific task, such as grading students, aligns with the intended application of the algorithmic advice. Contextual shifts may render the original algorithmic advice unsuitable, necessitating human judgment to optimize forecast potential in light of these changes.

Such forecasting adjustments serve a dual purpose: Assessing the applicability of algorithmic advice in a given context and acting as a control mechanism for potential biases. Moreover, evidence suggests (Karelaia & Hogarth, 2008; Seifert et al., 2023) that including contextual information complicates forecasters' judgment processes. This highlights the urgent need for further research on the influence of contextual factors on forecasting (Seifert et al., 2023, p. 110).

In addition, Pronin and Hazel (2023) show that humans have a bias blind spot that "prevents people from working to correct their biases and leaves them to instead impute bias to others—especially those who disagree with them" (see Pronin & Hazel, 2023, p. 1). Their review did not link the bias blind spot to algorithm advice. We argue that this blind spot may also play a role in applying algorithm advice—for example, in tasks where algorithmic advice disagrees with an expert's advice or when the advice is unexpected. Hence, when people are biased toward algorithm advice, they may also avoid addressing these biases due to their blind spots, and apply them to the algorithmic advice offered, thus reducing its effectiveness. Expanding on this notion, it is plausible to consider that such interactions could ultimately influence the advice generated by AI, including LLMs such as ChatGPT, which learn from human inputs. This creates a feedback loop, potentially reintroducing human biases back into the algorithms' advice.

To summarize, human biases have the potential to influence forecasting, and algorithms might affect these processes. We contend that algorithms hold substantial promise for enhancing forecasts by mitigating forecasting errors stemming from these biases. Consequently, we identify a gap in the existing literature—a lack of comprehensive analysis of the role algorithms have in improving forecasting accuracy. Our review aims to bridge this gap by elucidating how algorithms can facilitate more accurate forecasts. A paradigm shift in the discourse is imperative to fully harness the potential of algorithmic advice. This shift must incorporate a human-centric perspective into discussions about algorithmic interactions, moving beyond merely examining the biases inherent in algorithmic sources. By integrating the human decision-making aspect into the conversation, we can more effectively anticipate the future impact and applications of algorithmic advice, paving the way for advancements in this field.

2.3.2. Forecasting task types

Although numerous studies of algorithm use are summarized in several recent overviews (Burton et al., 2020; Jussupow et al., 2020; Kaufmann et al., 2023; Mahmud et al., 2022), none of these overviews focus explicitly on forecasting tasks. Although Kaufmann et al. (2023) differentiate between types of task, such as diagnosis and forecasting tasks, they do not present any in-depth analyses. Still, they provide an example of the two different types of judgments. For example, when teachers decide if students need extra tutorials, this is classified as a diagnosis tass (see Kaufmann & Budescu, 2020). A typical prediction task in their study sample is predicting a stock's value (see Chacon et al., 2022; Önkal et al., 2009).

A forecasting task usually involves predicting a future event. In Kaufmann et al.'s (2023) overall study sample, one-third of the tasks were forecasting tasks (28.7%, $k = 35$). In summary, we contend that forecasting tasks often

[3] See: https://press.un.org/en/2024/ga12588.doc.htm.

present greater complexity than diagnostic tasks, primarily due to the higher level of uncertainty inherent in the forecasting process. Therefore, in the present review, we differentiate between forecasting and diagnosis tasks in our analyses.

2.4. Our overview goals

Our overview has two goals: first, to analyze the potential effect of judgmental biases in algorithm use on forecasting tasks. Due to the missing focus on algorithms' potential to reduce the possible adverse effects of human judgmental biases affecting forecasting, we add this perspective to the literature. We highlight the impact of human biases in decision makers' forecasts and provide an additional perspective on the benefits that algorithm use might have. Second, regarding the type of tasks, to consider whether they were forecasting tasks or not. In other reviews, this information was, for the most part, not present in previous study samples. Hence, our overview adds to the literature the need to focus on different types of tasks (e.g., forecasting or diagnosing) in which algorithm advice is available.

Algorithm advice is implemented in several forecasting domains, and research has shown its impact in reducing risks. On the other hand, within algorithm use research, several reviews show a dominance of algorithm advice non-use and algorithm aversion. Although biased algorithms are often the focus of algorithm advice research, human judgmental biases and the type of judgment task (forecasting) are seldom considered. Hence, we add to the literature (a) an analysis of the effects of algorithm use in forecasting considering judgmental biases affecting forecasting and (b) a study of whether the type of tasks, namely forecasting tasks, is considered in algorithm use studies. The overall goal is to reveal any research gaps and guide future research to improve the accuracy and reliability of forecasting.

3. Method

3.1. Study sample

As outlined, four recent reviews (Burton et al., 2020; Jussupow et al., 2020; Kaufmann et al., 2023; Mahmud et al., 2022) provide a valuable state of the art on algorithm use literature. The four overviews considered 214 studies. If we exclude any repeated studies, we find 145 studies. Only three studies (Arkes et al., 1986; Dietvorst et al., 2015; Prahl & Van Swol, 2017) were considered in each overview.

In addition, we include papers on algorithm use from forecasting journals. We adhered to the following procedure to select these papers. First, we revisited the keywords highlighted in the four recent reviews on algorithm use, identifying "algorithm aversion" and "algorithm appreciation" as predominant themes. Second, in March 2024, we conducted a targeted search for journals in the Web of Science and Scopus databases that explicitly include "forecasting" in their titles, due to the absence of forecasting as a distinct category within these databases. This search yielded several key publications, such as *Advances in Business and Management Forecasting*, *Forecasting*, *International Journal of Forecasting*, *Journal of Forecasting*, *Romanian Journal of Economic Forecasting*, *Technological Forecasting and Social Change*, and *Weather and Forecasting*. Subsequently, we employed Google Scholar to search these identified journals for the key terms "algorithm aversion" and "algorithm appreciation". This meticulous process led to the discovery of 18 pertinent papers. After cross-referencing these findings with our existing database, we added 17 unique papers.

Consequently, we compiled a total of 162 studies (refer to the study reference list in Appendix 3) that constitute our sample for the current overview. This encompasses all studies from the four previous reviews, as well as studies published in forecasting journals. For a detailed enumeration of the studies included in the reviews and those from forecasting journals, please refer to Appendix 4. The number of publications per five-year period can be seen in Fig. 1 and per year in Appendix 5.

3.2. Study coding and analyses

In recent years, meta-analyses and systematic reviews have become increasingly automatic processes (Kaufmann, 2024; Van Dinter et al., 2021; Wagner et al., 2022). Hence, we integrate this trend using a text-mining exploratory approach to support our study's coding procedure. Text mining has been introduced as a helpful supporting tool in systematic reviews (see Ananiadou et al., 2009; Thomas et al., 2011) and can potentially increase their objectivity.

First, we obtained the 162 papers in PDF format and cataloged each study, noting the first author's name and publication year. Subsequently, we employed a text-mining approach to compute the frequency of the twelve forecasting biases, as Makridakis and Bakas (2016) outlined within each study. The twelve forecasting biases, as delineated and ordered in Table 6 of Makridakis and Bakas (2016) and mentioned in Appendix 1, are a search for supportive evidence, inconsistency, conservatism, recency, availability, anchoring, illusory correlations, selective perception, regression effects, attribution of success and failure, optimism, and underestimating uncertainty.

For our text-mining analyses, we used the R programming language (Core Team, 2022) and the "stringr" (Wickham, 2022), "pdftools" (Ooms, 2023), and "openxlsx" (Schauberger & Walker, 2022) packages (Hvitfeldt & Silge, 2022). This automated process generated a comprehensive frequency table detailing the occurrence of terms for biases across the 162 papers analyzed (see the "Used the term" column in Table 1). In addition to the terms listed in Table 1, we added more general terms such as "heuristic" or "bias" to see how much the paper covered the topic. Hence, we added a control and backup strategy for our coding procedure to ensure that the terminology used did not influence the results.

Utilizing the generated frequency table as a guide, we collaboratively examined and discussed each paper associated with a specific bias. This involved exploring the context and depth with which each paper addressed the

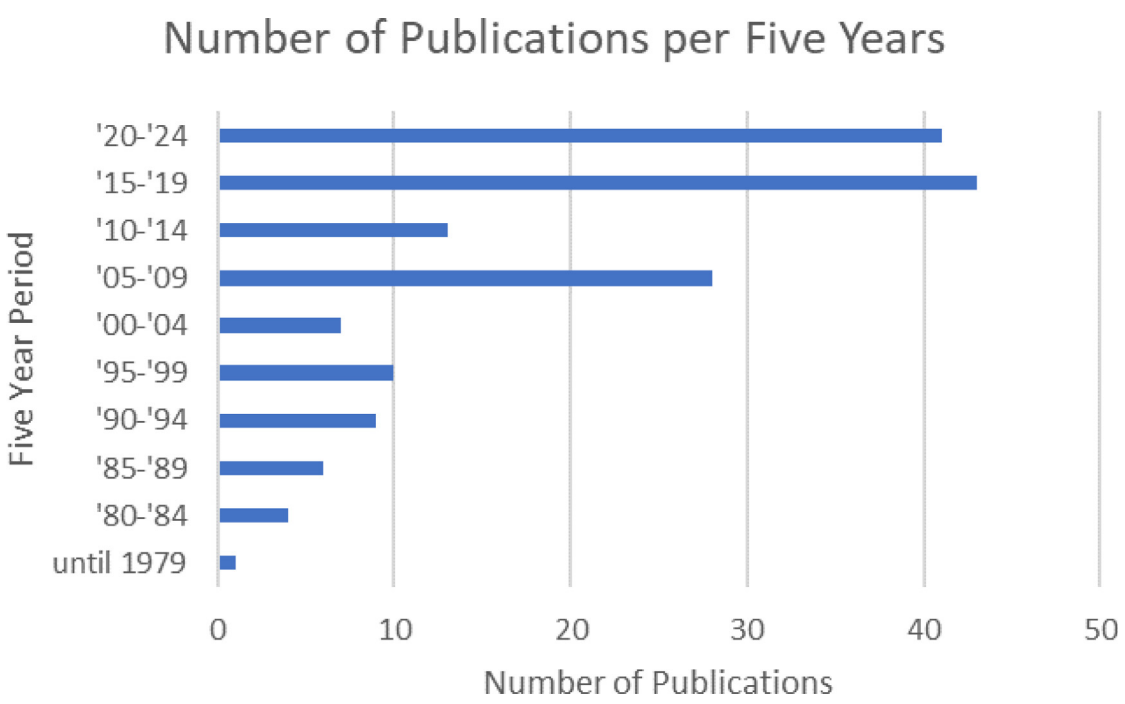


Fig. 1. Number of publications per five-year period in our sample of 162 studies.

Table 1
Summary of the different biases affecting forecasting in algorithm use studies ($N = 162$).

Bias	Used the term	In-depth analysis		Type of task (for effectively tackled)	
		Briefly mentioned	Effectively tackled	Forecasting	Diagnosis
Anchoring	23	9	5	3	2
Availability	57	7	2	2	0
Inconsistency	41	4	2	2	0
Confirmation bias	19	3	2	0	2
Recency	8	6	0	0	0
Optimism	17	5	0	0	0
Conservatism	7	1	1	1	0
Illusory correlations	1	1	0	0	0
Regression effects	1	1	0	0	0
Total (single counting)	102	25	8	5	3

Note. Some papers could be counted in more than one row, because more than one bias could be mentioned in a single paper. Therefore, "Total (single counting)" considers papers without duplication, ensuring each paper is counted only once.
"Effectively tackled" considers empirical papers that effectively addressed the topic of the corresponding bias.
"Type of task" considers only "effectively tackled" papers.

corresponding bias. Additionally, we verified the presence of each bias in every paper using the search function in the PDF reader. The results obtained through this manual verification process aligned with the outcomes the automated R script generated.

In addition to the text-mining approach, which helped us to reveal any biases mentioned by Makridakis and Bakas (2016) during our coding procedure, we coded the context of the mentioned biases. Each study analyzed was classified according to how deeply it covered the bias and whether it effectively addressed the topic or briefly mentioned it. We neglected papers that used the term in a manner unrelated to the corresponding bias (see the "In-depth analysis" column in Table 1). For the papers that effectively addressed the topic in empirical papers, we checked if the tasks employed were for forecasting or diagnosis, explaining the former first (see the "Type of task" column in Table 1).

In the analysis, we explore how the twelve most significant judgmental biases that impact forecasting are covered in the algorithm use literature. We grouped under the concept of confirmation bias "search for supportive evidence", and "selective perception", as described by Makridakis and Bakas (2016). This grouping makes sense because these biases can be seen as manifestations or forms of confirmation bias (Nickerson, 1998). It is worth noting that "attribution of success", "attribution of failure", and "understanding uncertainty" were not explicitly addressed in any of the 162 papers we reviewed.

The R script and Appendices for the present overview are available at the Open Science Framework (OSF): https://osf.io/38ypz.

4. Analysis and synthesis

In the upcoming subsections, we present the outcomes interpreted through the lens of our two research objectives: first, exploring the impact of algorithmic usage on biases in forecasting and second, elucidating the specific tasks associated with each identified bias. Each bias is discussed in its own subsection, with our arrangement based on the descending order of the total number of papers addressing that particular bias (combining "Briefly mentioned" and "Effectively tackled" categories). In each subsection, we begin by defining the bias, followed by a detailed examination of the findings from each study related to that bias. In our discussion of the first four critical biases—anchoring, availability, inconsistency, and confirmation—which were more addressed than other biases in the literature on algorithm use, we include a concluding paragraph summarizing our findings and offering recommendations. For the subsequent five biases, our discussion and recommendations are presented more succinctly in the *General Discussion and Recommendations* section. We consider the types of tasks explored in empirical papers and the depth of coverage they provided. A summary of the papers analyzed is shown in Table 1.

Table 1 provides an overview of the 162 papers analyzed. It reveals that 102 of these papers employ the terms, yet without associating them with the context of judgmental bias affecting forecasting. Only 25 papers, accounting for 15.4% of the total, make a brief or indirect reference to at least one of the biases highlighted by Makridakis and Bakas (2016). Even more scarcely, only eight papers, representing 4.9% of the total, address any of the identified biases directly. Among these eight papers, forecasting tasks were the focus in five, while the remaining three concentrated on diagnostic tasks.

4.1. Anchoring

Anchoring bias (Tversky & Kahneman, 1974) is a cognitive bias where individuals rely too heavily on the initial information (the "anchor") when making decisions or judgments. This initial anchor can influence their subsequent thinking and lead to systematic errors, even if the anchor is irrelevant or arbitrary. People often adjust their judgments away from the anchor insufficiently, causing them to be influenced by the initial information. Of the 162 papers, 23 included "anchoring" at least once. Conducting an in-depth analysis of these 23 papers, we identified three studies that effectively tackle the issue of judgmental anchoring bias in forecasting tasks, two studies that do so in diagnosis tasks, and nine studies that briefly mention the topic, as outlined below (see Table 1).

The oldest publication on forecasting tasks and bias in algorithm use is the study by Lim and O'Connor (1996). Their study is in the context of decision-making through information processing, where they assess the effectiveness of employing a decision support system (DSS). Regardless of the information's reliability, people depend heavily on their intuition when forecasting and thus do not acquire enough information. Consequently, they often fail to gather sufficient information, resulting in lower forecasting accuracy compared with situations where high-quality information was utilized. This conservative approach to information acquisition is problematic, as it is closely associated with anchoring in initial forecasts. Lim and O'Connor (1996) propose potential solutions to address the challenges posed by judgmental anchoring bias. They suggest reducing dependence on unaided forecasters or incorporating internal debiasing mechanisms within DSS as strategies to mitigate the negative effects of this bias.

Feng and Gao (2020) also analyze forecasting tasks and algorithm advice use. They conducted lab experiments to examine the influence of DSS recommendations on individuals' behavior. The authors posit that various behavioral biases, such as anchoring and insufficient adjustment bias, influence the pull-to-center bias. While the absolute optimality of a DSS recommendation may remain uncertain, Feng and Gao argue that these recommendations serve as anchor points in the decision-making process. Therefore, through the programming of DSS advice, decision makers are more likely to make decisions that align more closely to it due to insufficient adjustment. Consequently, the emphasis has shifted toward enhancing the quality of DSS advice, ensuring that the guidance is a suitable anchor that helps improve the forecast.

Finally, Van Dongen and Van Maanen (2013) introduce a framework for understanding tasks and psychological factors that influence one's propensity to rely on advice provided by decision aids. Their experiment yielded a discovery. Participants, despite perceiving decision aids as potentially more dependable than themselves, did not exhibit heightened reliance on these aids. The authors posit that this incongruity can be elucidated by considering factors influencing automatic and effortless cognitive processes, such as heuristic availability and anchoring bias. Related to anchoring bias, the authors propose that individuals are more inclined to invest the necessary mental effort to forsake their initial decision (anchor) and embrace alternative advice from a more dependable decision aid when they feel greater responsibility for the task's outcome.

As outlined before, there are two studies with diagnosis tasks that study algorithms and anchoring biases. Hertz and Wiese (2019) investigate the willingness to consider advice from nonhuman agents in a social and analytical task. In their studies, participants first chose the adviser and then the task. Allowing participants to select their advisers before receiving recommendations might have introduced the anchoring cognitive bias. This phenomenon could possibly elucidate why, in Hertz and Wiese's (2019) studies, the alignment between the agent and the task does not impact compliance with post-decision outcomes, while the initial commitment to a specific adviser does. Allowing people to choose their advisor initially could have the effect of their over-relying on the initial information provided. Brown and Jones (1998) mention the anchoring bias, developing a model of decision aid reliance considering a diagnosis task, where a partial decision strategy would be to use the anchor-and-adjust strategy in choice tasks.

Other studies briefly mention strategies employed in their experiments to reduce possible anchoring bias

(Fenneman et al., 2021; Goodyear et al., 2017, 2016; Litterscheidt & Streich, 2020; Longoni & Cian, 2022; Madhavan & Wiegmann, 2007). For example, Goodyear et al. (2017, 2016) note that in their experiments, participants received advice before making their decisions as a preventive measure against cognitive anchoring bias. Similarly, Sutherland et al. (2016) draw attention to the relevance of task-related and environmental factors, where pre-advice self-generated values might influence decision makers through anchoring bias. Finally, in Sage's (1981) literature report, anchoring bias is mentioned as a factor to consider when designing a decision support system, and in Benbasat and Taylor's (1982) literature report, anchoring bias is mentioned as a behavioral bias that affects probability estimates.

Anchoring bias significantly influences decision makers in different contexts (Bystranowski et al., 2021; Saposnik et al., 2016), and our analysis reveals that this theme has been somewhat explored within the literature on algorithmic use. Algorithms have the potential to play a central role in providing appropriate anchors, particularly when integrated into the initial forecasting stage, as this can lead to reduced subsequent adjustments by decision makers. Therefore, appropriate advice from algorithms should precede human judgment, enabling us to leverage anchoring bias beneficially. This method stems from the insight that decision makers tend to make only minimal modifications to initial advice. By strategically positioning this concept, anchoring bias, often perceived as a cognitive flaw, can be re-envisioned and employed to enhance decision-making in forecasting. Consequently, it is essential to assess individuals' initial choice of advisor thoughtfully to counteract the potential over-reliance on information tainted by anchoring bias.

4.2. Availability

The availability heuristic is a cognitive shortcut that leans on the readily accessible examples that surface in an individual's mind when they assess a particular subject, idea, approach, or choice (Tversky & Kahneman, 1973). This heuristic is founded on the belief that if something can be remembered easily, it must carry significance, or at the very least, greater significance than other solutions that are less effortlessly remembered. Of the 162 papers, 57 include "availability" at least once. After conducting an in-depth analysis of these 57 papers, we identified two studies that effectively tackle the availability heuristic in forecasting tasks and seven studies that briefly mention the topic, as outlined below (see Table 1).

Considering forecasting tasks, as previously mentioned in the *Anchoring* subsection, Van Dongen and Van Maanen (2013) present a framework that offers insight into the task-related and psychological elements that shape an individual's inclination to depend on advice given by decision support tools. The authors put forth a series of recommendations for designing and managing decision aids to mitigate biases. One such recommendation involves mitigating the information gap between the decision maker and the decision aid while offering performance feedback. A particularly pertinent suggestion aimed at diminishing the influence of the availability heuristic is to provide advice after individuals have already formed their initial opinions rather than before. If advice is delivered after an opinion has been formed, individuals may gain confidence when it aligns with the decision aid, while it may prompt them to reconsider their opinions when there is disagreement.

In another forecasting task, Whitecotton (1996) investigates the combined influence of experience and decision aids. Interestingly, the author discovered that participants with more experience (specifically, financial analysts in the experiment) exhibit higher bias levels, yet those participants achieve the highest accuracy levels overall. One explanation for this heightened bias is associated with availability bias: Financial analysts recall information from their past experiences that differs from the information provided in the experiment.

In their 2008 commentary paper, Phillips and Gully emphasize the idea that when the analytical approach functions effectively, it reduces the likelihood of the availability heuristic coming into play. Conversely, they also mention that algorithm errors tend to stand out and be remembered more readily, resulting in overestimation of the actual frequency of such errors. Additionally, the authors point out that in the absence of objective feedback, managers tend to maintain a high level of confidence in their expertise, a confidence that is influenced by the availability heuristic. Swinney (1999) suggests in her recommendations for future research that exploring factors that could reduce over- or under-reliance on expert systems is essential. Swinney (1999) mentions that experienced decision makers may be less susceptible to judgmental biases such as the availability heuristic.

In the literature report by Benbasat and Taylor (1982), availability is discussed as a behavioral bias that impacts the estimation of probabilities. Patterson's (2017) literature report cites the availability heuristic as an instance of simplifying heuristics that contribute to intuitive judgments. Sage's (1981) literature report highlights the availability heuristic as a factor to consider when designing a decision support system. Finally, Liu et al. (2019) and Mosier and Fischer (2017) mention the availability heuristic as an illustrative example of how individuals employ heuristics to shape their judgments and as factors that might produce incidents.

In conclusion, we recognize the importance of the availability heuristic as a crucial factor shaping individuals' decisions. This concept has been considered only partially in the literature related to algorithm use. Algorithms, when designed thoughtfully, have the capacity to harness the availability heuristic in a constructive manner. By offering relevant and timely cues, algorithms can create vivid and easily memorable information for forecasters. As with anchoring bias, algorithmic advice can exploit availability bias to enhance forecasting. This is because decision makers, upon receiving well-considered algorithmic advice, will likely be swayed by it if it evokes vivid memories. This influence increases their propensity to seek and value further advice from the same algorithmic source.

It is important to acknowledge a significant caveat: Algorithms may prompt changes even when the information they highlight has little or no predictive value. This can occur when the information is noticeable but not diagnostically useful (Fildes, Goodwin, & De Baets, 2024; Fildes et al., 2019). Additionally, there is a risk that the information provided by the algorithm may be misinterpreted, which raises concerns about expanding the set of information cues. As Fildes, Goodwin, and De Baets (2024) point out, software providers often increase the volume and complexity of information available to forecasters, which could be counterproductive. We concur with Fildes, Goodwin, and De Baets (2024) in noting that a key challenge will be to design organizational processes and algorithms that guide forecasters toward focusing on new information that is truly relevant to future demand.

4.3. Inconsistency

Inconsistency is failure to employ the same decision-making standards when faced with comparable circumstances (Makridakis & Bakas, 2016). The prevalence of inconsistency is one of the primary reasons individuals are frequently surpassed by straightforward linear models in various contexts (Dawes et al., 1989). In the context of forecasting, inconsistency can lead to issues such as inaccurate predictions, reduced reliability, and confusion. Forty-one of the 162 papers include the term "inconsistency" or "inconsistent" at least once. After conducting an in-depth analysis of these 41 papers, we identified two studies that effectively tackle the issue of inconsistency in forecasting tasks and four studies that provide a brief mention of the topic, as outlined below (see also Table 1).

As noted earlier in the *Availability* subsection, Whitecotton (1996) examines the joint impact of experience and a decision aid in a forecasting task. Notably, the study reveals that the decision aid decreases the scatter in forecasting accuracy, although this effect is not uniform across groups with different experience levels. Since consistency is manifested through the scatter component of accuracy, employing a decision aid could enhance forecasting accuracy by minimizing judgment variabilities.

In forecasting tasks, Ashton et al. (1994) examine decision-aiding techniques for advertising sales forecasts. Their findings indicate that all decision-aiding methods yielded more precise forecasts than those generated by managers alone. The authors observed that these models demonstrated perfect consistency, and over time, this consistency translated into significantly improved accuracy.

Some studies briefly mention the inconsistency concept. For example, Arkes et al. (1986) observe that physicians swiftly return to their prior, less accurate diagnostic performance once a computer system is discontinued, showing people's inconsistent decision-making. Similarly, Pagano et al. (2016) highlighted the dominance of models over people, which is commonly attributed to human inconsistency. In the same context, Goodwin et al. (2022) highlights, using the U.S. president Jimmy Carter as an example, the frequent complaints he made regarding the inconsistency of forecasts provided by economic advisors. In addition, Westin et al. (2016) emphasize the importance of taking into account individual differences, especially in internal consistency, in problem-solving styles when designing systems.

In summary, we acknowledge the significance of inconsistency as a pivotal element influencing individual decisions. The literature concerning algorithm utilization has explored inconsistency only to a limited extent. However, there is a consensus that algorithms surpass human performance by providing consistent advice over time. Algorithms exhibit greater stability and consistency when dealing with similar contexts and situations. Thus, algorithms could prove immensely valuable in enhancing the decision-making processes of individuals prone to inconsistency.

4.4. Confirmation bias: The search for supportive evidence and selective perception

Confirmation bias is a cognitive bias that refers to the tendency of individuals to favor, seek out, interpret, and remember information or evidence in a way that confirms or supports their preexisting beliefs or hypotheses while ignoring or downplaying contradictory information (Wason, 1960). A case study reported by Fildes and Goodwin (2021) illustrates how confirmation bias in algorithm use can lead to adverse outcomes if not carefully managed. In this instance, managers modify a system to align with their collaborative tasks and shared market perceptions, effectively embedding their views into the system and disregarding the original intentions of its designers. Confirmation bias can manifest in various forms, and while there is not a strict categorization, two of the most relevant forms of confirmation bias are the search for supportive evidence and selective perception. Out of the 162 papers, none make use of the term "search for supportive evidence", one uses the term "selective perception", and 18 include the term "confirmation" at least once. After an in-depth analysis of these 19 papers, none of the studies effectively tackles the issue of confirmation bias in forecasting tasks, two studies do so in diagnosis tasks, and three studies briefly mentions the topic, as outlined below (see also Table 1).

In diagnosis tasks, Scherer et al. (2015) conducted three experiments involving participants who were tasked with making hypothetical medical decisions. In their second study, the researchers aimed to understand whether patients' increased confidence stems from the effort put into decision-making or from confirmation bias. Their findings reveal that the rise in confidence is primarily influenced by confirmation bias rather than the effort invested in decision-making. The study's authors do not investigate the impact of algorithms or decision aids on mitigating biases in medical decisions.

As discussed earlier in the *Anchoring* subsection, Hertz and Wiese (2019) examine the impact of selecting an advisor at the outset in the context of diagnosis tasks. They contend that pre-committing to an information source before making a decision results in confirmation bias. This precommitment to initial choices leads individuals to place greater trust in their original selection, even in the presence of more reliable information.

Mosier and Fischer (2017) briefly mention confirmation bias concerning the level of domain expertise exhibited by decision makers. Phillips and Gully (2008) highlight the impact of confirmation biases on managers' utilization of decision aids. They observe that confirmation bias is likely to operate detrimentally, whereas any failure of the decision tool reinforces users' beliefs that the tool is ineffective. Similarly to various cognitive biases, Sage (1981) highlights the significance of different biases associated with selective perception in creating decision support systems.

In sum, while confirmation bias holds relevance in forecasting, it is evident that its exploration within the algorithm use literature remains incomplete, particularly concerning forecasting tasks, where there have been limited emphasis. While confirmation bias is typically viewed negatively, there are instances where it can be harnessed to enhance decision-making processes. For instance, nudging individuals to pre-select a well-performing algorithmic adviser can capitalize on confirmation bias. When people commit to such a choice beforehand, the inherent bias may work positively, ultimately leading to better outcomes.

4.5. Recency

Recency bias, also known as the recency effect, is a type of cognitive bias where individuals give more weight or importance to the most recent information acquired or the most recent experiences when making decisions or forming judgments (Makridakis & Bakas, 2016). In essence, people tend to remember and rely on recent events more than older ones, believing that recent events are more relevant or significant (Fildes, Goodwin, & De Baets, 2024). Eight of the 162 papers include the term "recency" at least once. After an in-depth analysis of these eight papers, none of the studies effectively tackle the issue of judgmental recency bias, and we identify six studies that briefly mention the topic, as outlined below (see also Table 1).

As with other cognitive biases, Sage (1981) highlights the importance of considering recency bias when designing decision support systems. Similarly, other researchers (Fildes & Goodwin, 2021; Fildes et al., 2019) highlight that forecasters often prioritize recent observations because of the substantial burden on memory of recalling numerous historical events and conditions. Consequently, despite the capability of algorithms and systems to identify long-term trends, decision makers, influenced by recency bias, frequently disregard the recommendations provided by these algorithms. Other studies briefly acknowledge the recency effect but do not focus on its specific implications for forecasting tasks or its interaction with algorithm usage. Instead, these studies refer to it primarily as a limitation or an explanation provided in their experiments (Dijkstra, 1999; Önkal et al., 2009; Swinney, 1999).

4.6. Optimism or wishful thinking

Optimism or wishful thinking bias is a cognitive bias that leads individuals to make overly optimistic predictions or forecasts about future events or outcomes (Makridakis & Bakas, 2016). In the context of forecasting, people tend to underestimate the likelihood of negative or undesirable outcomes and overestimate the probability of positive or desirable ones. This bias can affect the accuracy of predictions and forecasts, as it leads individuals to be overly optimistic in their assessments of future scenarios, often disregarding potential risks or unfavorable factors that could impact the outcomes they are forecasting. Seventeen of the 162 papers include the terms "optimism" or "wishful thinking" at least once. After an in-depth analysis of these seventeen papers, none of the studies address the issue of optimism or the inclination toward wishful thinking adequately. We identified five studies that cover the topic briefly, as outlined below (see also Table 1).

In certain fields and case studies, it has been observed that judgmental adjustments are prone to optimism bias (Fildes & Goodwin, 2021; Fildes et al., 2019). As a result, forecasts are often revised upward without sufficient evidence to support such decisions. Sanders and Manrodt (2003) emphasize the information inherent in processing limitations and biases in human decision-making, such as optimism and wishful thinking. Consequently, the reliance on human judgment in forecasting is discouraged due to its susceptibility to these shortcomings. As with other cognitive biases, in Sage's (1981) literature report, the author highlights the significance of considering optimism or wishful thinking bias when designing decision support systems. Related to this is the phenomenon of unrealistic expectations regarding forecasting advisers, whether human or models (Dietvorst et al., 2015). It can be problematic to anticipate an impossibly high level of performance from forecasters.

4.7. Conservatism

Conservatism bias refers to the tendency of individuals to cling to their existing beliefs or initial judgments while undervaluing new, contradictory, or updated information (Makridakis & Bakas, 2016). This cognitive bias can lead people to maintain outdated views or interpretations even when presented with compelling evidence to the contrary. In essence, conservatism bias reflects a reluctance to revise one's beliefs or opinions based on new and relevant information, which can hinder the accuracy and adaptability of decision-making processes. The conservatism bias often aligns with anchoring and confirmation bias, arguably serving as a means to lessen the cognitive effort required for processing additional data. Seven of the 162 papers included the term "conservatism" at least once. After an in-depth analysis of these seven papers, we identified only one study that addresses conservatism bias in forecasting tasks and another that briefly mentions the topic, as outlined below (see also Table 1).

As mentioned earlier in the anchoring subsection, Lim and O'Connor (1996) investigate conservatism bias. They emphasize in their study that individuals frequently rely on their intuition for forecasting and are inclined not to gather extensive information. Interestingly, regardless of the reliability condition, they observe conservatism bias in their experiment and note that conservative behavior in information acquisition led to inferior forecasting

accuracy and proved challenging to overcome. Much as with other cognitive biases, in his literature report, Sage (1981) briefly emphasizes the importance of considering conservatism when designing decision support systems.

4.8. Illusory correlations

Illusory correlations are the perception of a statistical relationship or correlation between two variables when, in reality, there is little to no actual connection between them (Makridakis & Bakas, 2016). People tend to perceive these false correlations when they believe that certain events or traits are associated with each other, even if there is no empirical evidence to support such a relationship. Among the 162 papers, only one references the term "illusory correlations" (see Table 1). Kuncel (2008) briefly touches upon the topic and briefly highlights the importance of training individuals to filter out irrelevant information during judgment, as this can potentially diminish decision makers' propensity to develop illusory correlations. Additionally, the author observes that certain decision-making rules can exhibit remarkable resistance to change, even when confronted with compelling evidence.

4.9. Regression effects

As Makridakis and Bakas (2016) notice, the presence of regression effects happens when sustained rises or falls are attributed to random factors. If this holds true, it will heighten the likelihood of a subsequent decline or increase in the opposite direction. Of the 162 papers, only one mentions "regression effects" (see Table 1). Sage's (1981) literature report employs the term, and as he notices, the presence of regression effects occurs when high recorded values of observations are employed without a regression to the mean to account for the impact of noisy measurements.

5. General discussion and recommendations

Our overview shows the importance of focusing not only on biases in algorithms but also on human biases in the human–algorithm interaction. Hence, it supplements current research on judgmental biases affecting forecasting in algorithm use. There is a noticeable research gap in the algorithm use literature concerning this challenge. Among the 162 papers analyzed in the four reviews (Burton et al., 2020; Jussupow et al., 2020; Kaufmann et al., 2023; Mahmud et al., 2022) and studies published in forecasting journals focusing on the utilization of algorithms, only five empirical papers (3.1%) with forecasting tasks addressed the effects of at least one of the twelve judgmental biases in forecasting identified by Makridakis and Bakas (2016). Three empirical papers (1.9%) considered diagnosis tasks, and 25 papers (15.4%) briefly or indirectly mentioned at least one of the twelve judgmental biases affecting forecasting. Despite this limited coverage, intriguing findings have emerged, and valuable avenues for future research are identified. When strategically integrated into the forecasting process, algorithms offer promising solutions to improve forecasting accuracy considering judgmental biases.

Among the biases examined, four are explored in the algorithm use literature: anchoring, availability, inconsistency, and confirmation bias. First, in line with the recommendations of Lim and O'Connor (1996) and Feng and Gao (2020), to reduce reliance on unaided forecasters and use DSS recommendations as anchoring points, we propose utilizing algorithms to provide accurate, objective, and data-driven reference points at the outset of the forecasting process. This approach can minimize dependence on skewed information. Anchoring bias can be used strategically to our advantage, improving decision-making in forecasting. Second, taking into account the availability heuristic, Van Dongen and Van Maanen (2013) and Whitecotton (1996), emphasize the importance of providing timely and relevant information. By doing so, algorithms can assist forecasters in accessing a wider range of data in a timely manner, overcoming the constraints of human memory and perception.

Continuing with the four biases explored in the literature on algorithm use, the third point concerns inconsistency. While human decision-making may fluctuate over time, algorithms provide consistent responses to similar situations. As Ashton et al. (1994) and Whitecotton (1996) note, by incorporating algorithms, decision makers can benefit from the stability and reliability of these systems, promoting more consistent and accurate forecasting outcomes. Last, confirmation bias can be harnessed positively in algorithm use despite being underexplored. For example, Hertz and Wiese's (2019) pre-commitment strategy encourages people to choose a high-performing algorithmic advisor initially, which can leverage confirmation bias to advantage.

The remaining biases, including recency, optimism/wishful thinking, conservatism, illusory correlations, regression effects, attribution of success and failure, and understanding uncertainty, as outlined by Makridakis and Bakas (2016), were either briefly addressed or, in some cases, entirely overlooked. Among the biases briefly mentioned in the algorithm use literature, we first notice recency bias. Algorithms, particularly statistical models, help mitigate recency bias by incorporating a broader range of data, including historical trends and averages, thus reducing the influence of any one recent observation (Fildes et al., 2019). Moreover, the use of structured forecasting support systems can help filter and integrate qualitative and quantitative information, ensuring that judgmental adjustments are made based on a comprehensive analysis rather than the most salient recent data. These systems can include features that adjust automatically for potential biases, offering suggestions that counteract the human tendency to focus too heavily on the latest data (Fildes & Goodwin, 2021). Concerning optimism/wishful thinking, algorithms can reduce their impact on judgmental forecasting by providing objective, data-driven insights (Sanders & Manrodt, 2003). Additionally, combining algorithms with structured judgment may improve forecast accuracy by mitigating optimism (Fildes et al., 2019).

Continuing with the remaining briefly mentioned biases, algorithms' ability to provide a broader range of

information and recommendations can counteract conservatism bias. This, in turn, enables a more balanced and informed decision-making process, potentially leading to improvements in forecasting accuracy (Lim & O'Connor, 1996). In the case of illusory correlation bias, structured methods can prevent faulty judgment that arises from perceived but non-existent patterns (Kuncel, 2008). Therefore, algorithms can mitigate the negative impact of illusory correlations by enforcing more objective decision-making processes. Finally, algorithms can help reduce the impact of regression effects in judgmental forecasting by enforcing a more statistically sound approach that counteracts human tendencies to overlook uncertainties and overly rely on extreme values (Sage, 1981).

After the empirical papers that utilize forecasting and diagnosis tasks are analyzed, it becomes apparent that only eight tackle biases effectively. Among these, five papers focus on forecasting tasks, while three center on diagnosis tasks. Despite the limited sample size, this observation highlights the significance of forecasting tasks in the algorithm use literature. We advocate continued research in this area, emphasizing the importance of forecasting in shaping judgments and decision-making processes, particularly in conjunction with algorithms. As Fildes, Goodwin, and De Baets (2024) mention, it is important to recognize the limitations of the information that algorithms can process, particularly when certain data are unavailable to them. The judgmental adjustments made by humans to address these gaps can introduce flaws. In addition, as mentioned earlier, LLMs such as ChatGPT learn from human input. This interaction can influence the advice provided by AI, potentially creating a cycle where human biases are reinforced in the AI's responses.

Our overview shows the lack of studies that consider judgmental biases affecting forecasting in algorithm use research properly, especially in forecasting tasks. Although some biases are considered in the literature more commonly than others, we suggest that future studies should consider developing/evaluating algorithmic decision aids that engage a wider range of psychological heuristics and biases. It is important to note that the term "bias" has multiple interpretations (Hahn & Harris, 2014, p. 42) and is not always viewed negatively. In certain aspects of our lives, biases can be beneficial for survival, such as avoiding fire or food that smells bad; these are known as adaptive biases. Therefore, further research could investigate forecasting tasks where adaptive biases can enhance the value of forecasts or where it is necessary to adapt algorithmic advice accordingly.

While we focus on twelve specific biases, critics might argue for a more comprehensive approach. However, it is essential to emphasize that these twelve biases are the most relevant biases affecting forecasting (Makridakis & Bakas, 2016), and other bias lists or taxonomies were analyzed (Acciarini et al., 2020; Barnes, 1984; Jussim, 2012; Krueger & Funder, 2004; Oeberst & Imhoff, 2023; Pronin & Hazel, 2023). Only Makridakis and Bakas (2016) provide a list of biases with a specific focus on forecasting contexts, making their list the most suitable for our analysis (see Appendix 2). Thus, we conclude that the twelve most relevant judgmental biases affecting forecasting, as mentioned by Makridakis and Bakas (2016), offer valuable insights into the potential biases researchers might encounter when employing algorithms for forecasting.

Additionally, we want to emphasize the significance of our text-mining approach in our coding process. As previously mentioned, the automation of tasks has become increasingly vital in systematic reviews and meta-analyses. This trend helps mitigate human errors, especially given the ever-growing volume of publications that require thorough scrutiny. Through our text-mining approach and revision of results, we slightly increased the workload and significantly enhanced the quality of our analysis. Consequently, this approach bolstered our findings' reliability and our coding procedure's objectivity.

Our overview demonstrates that the challenges in forecasting tasks stem from algorithm biases and human biases in algorithm use. We noticed a gap in research concerning biased human decision-making and its impact on the feedback loop involved in developing and using algorithmic advice. Future studies should be initiated to address this gap. Consequently, our call for further research on this topic may pave the way for more extensive future reviews or a comprehensive meta-analysis (Hunter & Schmidt, 2004; Kaufmann, 2024). Such efforts could offer deeper insights into the role of algorithms in forecasting, particularly in mitigating judgmental biases, and enhance our understanding of how to leverage algorithmic advice to improve forecasting accuracy.

5.1. Practical implications

Our analysis reveals several actionable insights. Below, we highlight three key practical implications we believe are particularly relevant. First, practitioners aiming to utilize algorithms for reducing judgmental biases in forecasting should first recognize the value of integrating algorithmic insights into the initial stages of the forecasting process (see also Burton et al., 2020, 2023). By doing so, they can leverage objective, data-driven advice to mitigate the influence of skewed anchors. This strategy involves the prioritization of algorithm-generated forecasts to set a neutral, objective baseline before human judgment comes into play. By grounding decisions in data rather than intuition or initial impressions, organizations can improve the accuracy of their forecasts and decision-making processes. For instance, in financial forecasting, incorporating algorithmic models that analyze historical data and market trends can help analysts form more accurate initial estimates, reducing the influence of potentially skewed prior information (Jia, 2021). However, we should be mindful that when provided with additional information, people may adjust their forecasts by focusing too heavily on model-based anchors, potentially overlooking other valuable insights (Sroginis et al., 2023). This can ultimately harm the accuracy of their forecasts.

Second, to combat the limitations imposed by the availability heuristic and human memory, practitioners are advised to implement algorithms—whether simple or complex—that incorporate a broad range of relevant and timely information. This approach ensures that forecasters are not just relying on the most readily available

data or recent experiences but are considering a broader spectrum of information. Algorithms can sift through vast datasets to highlight critical insights that might otherwise be overlooked, encouraging a more comprehensive analysis that counters the natural human tendency to myopia and selective attention. For example, in supply chain management, predictive algorithms can analyze a variety of factors beyond recent trends, such as long-term consumer behavior patterns and global market dynamics, to provide more balanced advice (Kumar & Sharma, 2023).

Finally, practitioners should deploy algorithms as a means to achieve stable and reliable forecasting outputs. The inherent consistency of algorithms, as opposed to the fluctuating nature of human judgment, ensures that similar situations are assessed based on a steady set of criteria over time. This consistency not only enhances the reliability of forecasting outcomes but also builds trust in the process, as stakeholders can be confident that decisions are not swayed by changing human biases or external pressures. By embedding algorithms into the decision-making framework, practitioners can foster a culture of accuracy and dependability in forecasting endeavors. In sectors such as retail, where predicting consumer trends is crucial, algorithms can offer consistent advice that helps avoid the pitfall of confirmation bias, ensuring that decisions are based on comprehensive data analysis rather than selective evidence (Ferreira et al., 2016).

These practical recommendations outline how leveraging algorithms can play a pivotal role in reducing judgmental biases affecting forecasting. By incorporating these strategies, practitioners can significantly improve the accuracy and reliability of their forecasting efforts, ultimately leading to more informed decision-making processes. We must keep in mind the importance of exercising caution during the implementation process, as the design and subsequent use of systems can unintentionally reinforce biases, potentially undermining our original objectives.

5.2. Outlook: Generative AI and LLMs development

Generative AI and LLMs are transforming the forecasting landscape by introducing more sophisticated methodologies. These advances, particularly in AI technologies like LLMs, have the potential to impact forecasting processes significantly (Trovero & Potamitis, 2023). LLMs have demonstrated positive predictive value and high specificity in various applications, such as human resource management (Deepa et al., 2024) and neurological complication risk assessment in spinal deformity surgery (Schonfeld et al., 2024). Studies have explored the development and utilization of LLMs in diverse fields, including multimodal time-series forecasting (Jia et al., 2024) and traffic accident forecasting (de Zarzà et al., 2023).

The success and impact of LLMs are increasingly recognized, with discussions on their implications across various aspects of life and work, including forecasting (Makridakis et al., 2023). However, it is essential to acknowledge and address judgmental biases inherent in forecasting processes, especially with the integration of generative AI and LLMs. As LLMs continue to evolve and demonstrate near-human- or above-human-level performance in various tasks, the integration of LLMs in forecasting shows promise. Current research suggests that the adverse impacts of judgmental biases affecting forecasting can be mitigated through the use of algorithms, or alternatively, these biases could be strategically harnessed to enhance forecasting. However, further research could focus on how different types of algorithms or the terminology employed to refer to algorithms (e.g., "algorithm", "AI", "LLMs") might impact decision makers' behavior or be better suited to address the various types of biases involved in forecasting.

CRediT authorship contribution statement

Alvaro Chacon: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Esther Kaufmann:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Conceptualization, Data curation, Formal analysis, Project administration, Supervision.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT to check some sentences in order to improve grammar and increase active voice. After using this tool/service, the authors reviewed and edited the content as needed and revised it with a professional language service. The authors take full responsibility for the content of the current manuscript.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors thank the Editors and the anonymous reviewers for their insightful and constructive comments. The authors contributed equally to this work.

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